

Session 31: Post Class tests

1. DBK Bank paid out \$ 80 million in dividends on net income of \$100 million in the most recent year. The book value of equity for the firm is \$800 million. Assuming that the bank maintains its current payout ratio and return on equity in perpetuity, what is the expected growth in earnings per share in perpetuity?
 - a. 8%
 - b. 2%
 - c. 5%
 - d. 2.5%
 - e. None of the above
2. Viaconda Inc. is a tourism company that reported \$10 million in net income on a book value of equity of \$110 million in the most recent year; the company generated \$ 1 million in after-tax interest income on a cash balance of \$20 million. The company also reported net capital expenditures of \$4 million, an increase in working capital of \$ 2 million and an increase in total debt of \$3 million during the year. Assuming that it maintains its current non-cash ROE and equity reinvestment rate, estimate the expected growth in non-cash net income in the future.
 - a. 3.33%
 - b. 6.67%
 - c. 3.33%
 - d. 6.00%
 - e. None of the above
3. Pokemon Inc. is a toy manufacturer that reported after-tax operating income of \$50 million in the most recent year. At the start of the year, the company reported book value of equity of \$400 million, book value of debt of \$250 million and a cash balance of \$150 million. The company also reported capital expenditures of \$75 million, depreciation of \$ 30 million and a decrease in non-cash working capital of \$5 million. Assuming that it plans to maintain its current return on invested capital and reinvestment rate, what is the expected growth in operating income?
 - a. 6.15%
 - b. 8.00%
 - c. 13.33%
 - d. 10.00%
 - e. None of the above
4. Nevis Enterprises reported a return on invested capital of 15% in the most recent year and a reinvestment rate of 60%. The firm expects its return on capital to rise to 18% over the next 5 years on both existing investments and new investments. What will the compounded average annual expected growth rate be over the five years?
 - a. 10.8%
 - b. 9%
 - c. 14.51%
 - d. 12.71%

- e. None of the above
5. You are forecasting the operating earnings for TalkMedia, a young, high growth social media company.

	Last year	1	2	3	4
Revenues	\$100	\$200	\$320	\$450	\$600
Operating Margin	-10%	-5%	0%	5%	10%
Operating Income	-\$10.00	-\$10.00	\$0.00	\$22.50	\$60.00

The firm currently has invested capital of \$50 million. If the sales-to-capital ratio is 2.00, what will the pre-tax return on capital be in year 4?

- a. 120%
- b. -20%
- c. +20%
- d. 10%
- e. None of the above

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1. **d. 2.5%.** The payout ratio is 80% (80/100) and the return on equity is 12.5% (100/800). The expected growth rate in earnings per share = (1- Payout ratio) (Return on equity) = (1-.8) (.125) = 2.5%.
2. **a. 3.33%.** First, compute the non-cash ROE and equity reinvestment rate
 - Non-cash Net Income= 10 -1 = \$9
 - Non-cash ROE = (10-1)/ (110-20) = 10%
 - Equity reinvestment rate = (4+2-3)/ 9 = 33.33%
 - Expected growth rate = 10% (.3333) = 3.33%
3. **b. 8%.** The keys are the reinvestment rate and return on invested capital:
 - ROIC = 50/ (400 +250-150) = 10%
 - Reinvestment Rate = (75-30-5)/50 = 80% (Decrease in WC reduces reinvestment)
 - Expected growth rate = 10% (.80) = 8%
4. **c. 14.51%.** Since the return on capital is changing on existing and new investments, there are two components to the expected growth:
 - Expected growth from new investments = .6*.18 = 10.8%
 - Expected efficiency growth over 5 years = (.18-.15)/.15 = 20%
 - Expected efficiency growth per year= $(1.20)^{1/5}-1 = 3.71\%$
 - Expected annual growth = 10.8% + 3.71% = 14.51%
5. **c. 20%.** The key is to estimate the reinvestment each year, based upon the change in revenues and the sales to capital ratio. That reinvestment adds to the invested capital each year:

	Last year	1	2	3	4
Revenues	\$100	\$200	\$320	\$450	\$600
Operating Margin	-10%	-5%	0%	5%	10%
Operating Income	-\$10.00	-\$10.00	\$0.00	\$22.50	\$60.00
Reinvestment		\$50.00	\$60.00	\$65.00	\$75.00
Invested Capital	\$50.00	\$100.00	\$160.00	\$225.00	\$300.00
Return on capital	-20%	-10%	0%	10%	20%